



**Politecnico
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Software Design and Architecture

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Capitolo 1

Introduction to Software Design

1.1 Complexity

Complexity is anything related to the structure of a software system that makes it hard to understand and modify the system.

$$C = \sum c_p t_p$$

Where C is determined by all software parts p , c_p is the complexity of part p , and t_p is the fraction of time spent on part p .

1.1.1 Syntax complexity

- Change amplification: change require operating in different places
- Cognitive load: how much information to learn to understand and perform a task. Some times is better to write longer code to reduce the complexity and so the cognitive load.
- Unknown unknowns: there are partes of the system that we don't know where they are or even that they exist.

1.1.2 Cause of complexity

Dependency: When we need to understand or modify a part of the code, we need to understand or modify other parts of the code. Try to reduce the number of dependencies and the complexity of the dependencies.

Obscuiry: alla the important information shoul be clearly visible.

1.1.3 Strategical vs. Tactical Programming

- **Tactical programming:** focus on having code that works as fast as possible. "Move fast and break things".
- **Strategical programming:** focus on long term maintainability and scalability. "Working code is not enough".

1.2 Modules

To improve and eliminate dependency and complexity, we can use modules. A module is a part of the code that has a well defined interface and a well defined implementation.

- **Interface:** what we need to know to use the module. Formal part, possible checked by the compiler. Represents a cost.

- **Implementation:** how the module works. Ideally much more complex than the interface. Represents a benefit.

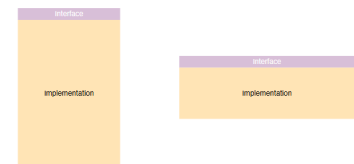
1.2.1 Abstraction

What we typically do is to perform an abstraction. Abstraction means selective removal of information. Is a simplified view of the something that omits "unimportant" information. Omitting information can lead to errors and keeping unimportant information can lead to cognitive overload.

Deep modules and shallow interfaces

Deep modules: modules with a simple interface and a complex implementation. They are good because they reduce the cognitive load and the change amplification. They are bad because they can lead to unknown unknowns.

Shallow interfaces: modules with a complex interface and a simple implementation. They are good because they reduce the unknown unknowns. They are bad because they can lead to cognitive overload and change amplification.



1.2.2 Information Hiding

Information hiding is the principle of hiding the implementation details of a module from the users of the module. Example sort function: we want to sort a list of numbers, we don't care about how the sorting algorithm used.

1.2.3 Information Leakage

Is the opposite of information hiding. Occurs when the implementation details of a module is reflected in multiple modules with consequences of creating dependencies.

1.2.4 Information Hiding Red Flags

- **temporal decomposition:** execution order is reflected in the code structure. Operations that happen in different times are in different classes.
- **overexposure:** API forces knowledge about rarely used features to use common ones. E.g.
`BufferedReader r = new BufferedReader(new InputStreamReader(System.in));`
- **Too many classes:** code is divided into too many shallow classes.

1.2.5 Method separation

Better together or Better apart?

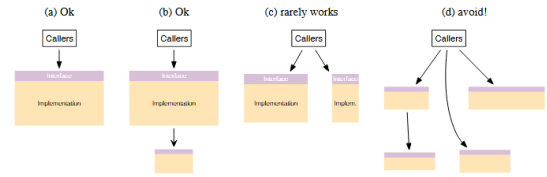
- **Bring together:** if information is shared, if we simplify the interface, if we eliminate duplication.
- **Separate:** if the code is general purpose or specific purpose.

Method separation red flags

If we have a lot of repeated code and/or we mixed specific and general code, means we have put together code that should be separated.

1.2.6 Do one simple thing and complexity

Each method should do one thing and do it completely



1.3 Obscurity

Obscurity is the principle of making the important information clearly visible and the unimportant information hidden. Obscurity is a great source of complexity. Obscure code is hard to Understand and creates more Bugs.

The solution to obscurity is writing obvious code, this means with a quick glance we can understand what the code does. We can achieve this by using abstraction layers.

1.3.1 Abstraction layers

Software systems are composed in layers. Higher layers use the facilities provided by lower layers.

Abstraction simplifies understanding and reduces obscurity.

In a well-designed system, each layer provides a different abstraction from the layers above and below it.

Abstraction Layers Red Flags

Some red flags that indicate that we have a problem with abstraction layers are:

- **Pass-Through methods:** is a method that does nothing but call another method.
- **Pass-Through variables:** is a variable that is passed through over and over down a long chain of method calls.

1.3.2 Writing Comments

Another method to reduce obscurity is writing comments. If users need to read a method code to use it, there is no abstraction.

Comments should describe things that are not obvious in the code:

- **Pick conventions:** write comments like other persons in the project
- **Don't repeat code:** avoid redundant comments that simply restate what the code already makes clear
- **Higher level comments enhance intuition**
- **Delayed comments are bad comments**
- **Use comments as design tool**

1.3.3 Naming

Toggether with comments, naming is a great tool to reduce obscurity. Bad names cause bugs. Name should create a image in mind of the reader about the nature of the thing being named. Names should be precise and consistent.

1.4 Errors and Anomalies

Exceptions add complexity because they create dependencies between the code and if they are easy to throw they are hard to handle.

A piece of code might throw exceptions in different ways:

- Caller may provide bad arguments or configuration
- Invoked method may not be able to complete the request
- In distributed systems network packets may be lost or delayed or peers may communicate in unexpected ways
- The code may detect bugs or internal inconsistencies or situations that are not prepared to handle

1.4.1 Define Errors out of Existence

What are the solutions to errors? Define errors out of existence. This means, not ignore the error but to redefine the behavior of a method so that no error is required.

E.g. `String.substring(int beginIndex, int endIndex)` it throws different exceptions in only specific cases. This is completely avoidable and unnecessary.

1.4.2 Limit exceptions

There are some methods to limit the number of exceptions that a method can throw:

- Handle exceptions at the point where they are thrown, if possible
- Exception aggregation means handle different exceptions with same handler
- Just crash
- Mask exceptions avoiding unexpected exceptions

1.5 Dependency Dimensions

1. Structural (Code-Level) Dependencies
2. Runtime / Distribution Dependencies
3. Data-Level Dependencies
4. Behavioral Dependencies

1.5.1 Structural (Code-Level) Dependencies

- **Implementation Dependency:** some piece of code depends on a concrete class instead of an abstraction
- **Construction Dependency:** you depend on how a collaboration, another class, is created.
- **Compile-Time Dependency:** the module knows something at compile time

1.5.2 Runtime / Distribution Dependencies

Often linked to distributed architectures.

- **Spatial Dependency:** A component depends on the location of another component. E.g. hard-coded URLs
- **Temporal Dependency:** A component depends on another being available at the same time. E.g. remote method calls or servers
- **Synchronization Dependency:** A component depends on another to complete work before continuing. E.g. synchronous request/response

1.5.3 Data-Level Dependencies

- **Data Dependency:** One component depends on another's internal data model. E.g. shared database
- **Schema Dependency:** Multiple services depend on the same database schema or representation structure. E.g. DB schema, XML/Json schema

1.5.4 Behavioral Dependencies

- **Control Flow Dependency:** One component dictates another's execution path. E.g. Centralized process managers
- **Transactional Dependency:** E.g. Components rely on atomic distributed consistency.
- **Timing Assumption Dependency:** Assumptions about response time or ordering. E.g. "Service must respond in <50ms" or Event ordering assumptions

Capitolo 2

Design Patterns

Initially proposed by Christopher Alexander in the context of architecture.

A reusable **solution**
to a known **problem**
in a well defined **context**

The context is a *design* situation giving rise to a (design) problem.

The problem is a set of forces repeatedly arising in the context.¹

The solution is a proven resolution of the problem. Configuration to balance forces. Structure with components and relationships and run-time behaviour.

Example of Social Pattern: You are in queue to the supermarket gastronomic section. The problem is hard to spot who arrived first. The solution is to use a ticket machine.

2.1 Types of Software Patterns

- **Architectural patterns** - high level structures of software systems. E.g., layered architecture, client-server, microservices.
- **Design patterns** - solutions to common design problems. E.g., Singleton, Observer, Factory Method.
- **Idioms** - language-specific patterns. E.g., RAII in C++, list comprehensions in Python.

2.1.1 Architectural pattern

Normally the architectural pattern works with big components and their interactions and is needed to describe the responsibilities of these components. Expresses a fundamental structural organization schema for software systems and defines the rules and guidelines for organizing the relationships between the components.

Example of architectural pattern: Several programs that are used in sequence read from input and write sequentially to output.

The problem is there are a lot of intermediate files used for communication between programs.

The solution is to adopt pipes and filters architecture feeding with the result of the previous one.

2.1.2 Design patterns

Provides a scheme for refining components of a software system or their relationships and describes a commonly recurring structure of communicating components.

¹Any relevant aspect of the problem E.g., requirements, constraints, desirable properties

Example of design pattern: A class library providing few functionalities contains a lot of classes. The problem is the user is exposed to the internal complexity of the library. The solution is to create a new façade class that interacts with the user and hide all the details.

2.1.3 Idioms

It is a low-level pattern specific to a programming language. Describes how to implement particular aspects of components or the relationships between them. Leverages the features of a programming language.

Example of idioms: An attribute is constant and should be globally available to many classes.

The problem is opening access would allow unauthorized modifications and the attribute is repeated in every object.

The solution is to make public static final in Java language.

2.2 Pattern language

Pattern do not exist in isolation in fact can happen that two or more patterns are applied together, used to implement part of another pattern and introduce a problem solved by another

They are called Pattern Languages or pattern systems.

Collection of patterns together with guidelines for: Should:

- Implementation
- Combination
- Practical use
- Count enough patterns
- Describe patterns uniformly
- Present relationships

2.3 Idioms

X-ability

Clonability, serializability, comparability, etc. some specific behaviour dealing with many heterogeneous classes. The solution is to derive from a class or interface that defines the required methods.

Example: In Java all methods extends the root class *Object* and provides the methods *toString()*, *equals()*, *hashCode()*, etc. that can be overridden to provide specific behaviour.

2.3.1 Strong typing vs. Duck Typing

"If it looks like a duck, swims like a duck, and quacks like a duck, then it probably is a duck."

Strong typed language: You must define interfaces and classes to be able to call a specific method. Compile time type checking.

Dynamic/duck typed language: You just add a method to a class and you can call it. No compile time type checking.

2.3.2 Property

A property, more general than an attribute, is a named characteristic of an object typically accessed through well-defined operations called accessors, getters, and setters forming the public interface of the object and it is not necessarily an attribute of the object (e.g. the age can be derived from the date of birth).

2.4 Design Patterns

Description of communicating objects and classes that are customized to solve a general design problem in a particular context.

A design pattern names, abstracts, and identifies the key aspects of a common design structure that make it useful for creating a reusable object- oriented design.

		Purpose		
		Creational	Structural	Behavioral
Scope	Class	1	1	2
	Object	4	6	10

2.4.1 Pattern selection

How to use patterns? First of all you need to know the problem you are trying to solve, knowing the patterns and then select the one that best fits your needs.

2.4.2 Creational patterns

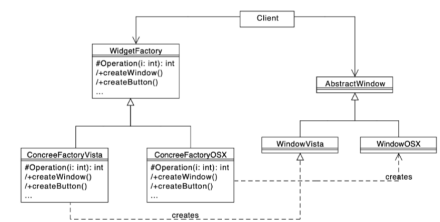
- Class
 - Factory Method
- Object
 - Abstract Factory
 - Builder
 - Prototype
 - Singleton

Abstract Factory

A family of related classes can have different implementation details.

The client should not know anything about which variant they are using / creating.

Example: GUI where you have different components (buttons, text fields, etc.) and the same GUI can be applied to different platforms such as Windows, MacOS, Linux, etc.

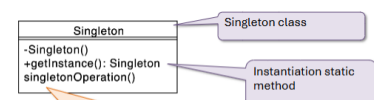


Singleton

A class represents a concept that requires a single instance.

The client should be able to access, in appropriate way, the unique instance.

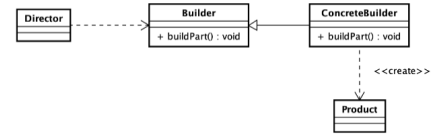
Note: If we talk about of an object designer pattern, in python you can just create a global variable and it will be a singleton.



```
private Singleton() { }
private static Singleton instance;
public static Singleton getInstance(){
    if(instance==null)
        instance = new Singleton();
    return instance;
}
```

Builder

An object of a complex class has to be created.
The creation entails complex interaction with the object and could be different variations of the object.



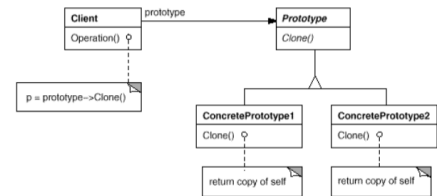
Example: $10.40 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-3}$

```
//Usual non-fluent builder
Measure power = new Measure(10.4);
power.addUnit("kg", 1);
power.addUnit("m", 2);
power.addUnit("s", -3);
power.setPrecision(2);
```

```
//Fluent builder
Measure power = new Measure.Builder(10.4)
    .is("kg")
    .by("m")
    .squared()
    .by("s")
    .to(-3)
    .withPrecision(2)
    .build();
```

Prototype

Several object can be created out of a palette of similar classes.
The class hierarchy can become quite complex and classes in hierarchy are differentiated by minor parameters (that could be defined through constructor arguments).
A solution could be start with a prototype classe and clone it to create new objects.



Example: A graphical editor where you have very similar shapes (e.g. square with different radius, colors, etc.).

2.4.3 Structural patterns

How organize classes and objects to compose larger structures.

- Class

Adapter

- Object

Bridge

Composite

Decorator

Facade

Flyweight

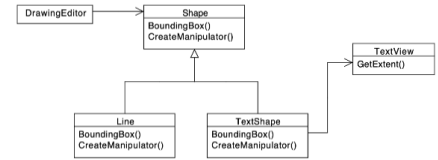
Proxy

Adapter

A class provides the features required by another class but its interface is not the one expected by the client.

The integration of the provider class should be possible without modifying it. The source code could be not available or is already used as it is somewhere else.

Example: In Java, keyboard events are handled by `KeyListener`. Listener class need to implement the interface and implement all the methods. If you are interested only in one of the methods, you can use the `KeyAdapter` class that is empty and you can decide which method to override, and the Listener extends `KeyAdapter` instead of `KeyListener`.

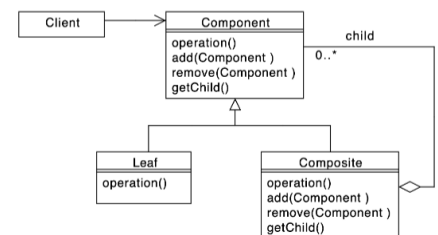


Composite

You need to represent part-whole hierarchies of objects.

Clients can be very complex and they require to adapt the difference between all the elements in the hierarchy.

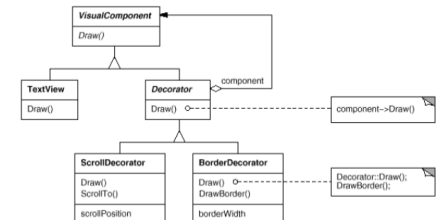
Example: In Java UI components are organized in a tree structure. You can have a panel that contains other panels and buttons, etc.



Decorator

You have some objects of classes that have specific behaviour and you want to add the same behaviour plus something else dynamically.

The solution can be extend the class but if that behaviour is something you would to apply to many classes, you need to extend all of leaf classes. Using a decorator you can modify the behaviour without creating an explosion of subclasses for every combination of features

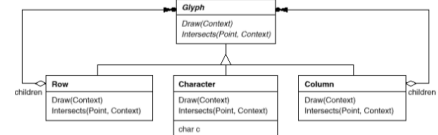


Example: In Java we have `JScrollPane` can be used to provide a scroll pane to any component

Flyweight

You need need to create a very large number of similar objects, and memory usage and performance become concerns.

Example: In Java we can pool different objects together. The `Integer` class in Java.



Proxy

You need to controll access to an object.

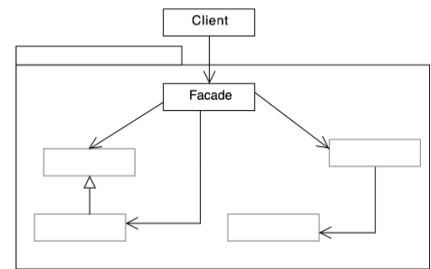
We need to provide a placeholder that is able to mediate between the client needs and all the requirements you want to controll before accessing the final object without changing interface or class.

- **Remote proxy:** provides a local representative for an object in a different address space.
- **Virtual proxy:** creates expensive objects on demand. E.g. slow loading resource - images in a web page.

- **Protection proxy:** controls access to the original object. E.g. when objects should have different access rights.
- **Smart reference:** is a replacement for a bare pointer that performs additional actions when an object is accessed. E.g. counting the number of references, checking permissions, etc.

Facade

Hide the details and the complexity of a general class by providing a simpler interface. The client should not know anything about the subsystem.



2.4.4 Behavioral patterns

Behavioral patterns allow to implement specific algorithms using objects and classes to provide control and flexibility.

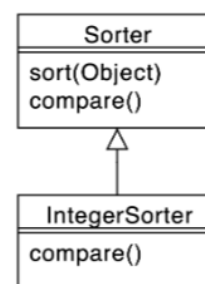
- Object
 - Chain of Responsibility
 - Command
 - Iterator
 - Mediator
 - Memento
 - Observer
 - State
 - Strategy
 - Visitor
- Class
 - Interpreter
 - Template Method

The basic mechanism of behavioral patterns is encapsulation variation. Instead of having a lot of if-else you have a structure at object level that allows you to define different alternatives.

2.4.5 Template Method

An algorithm/behavior has a stable core and several variation at given points. You have to implement/maintain several almost identical pieces of code.

Example: The compare in Java. You have to implement the compare method that solves integer or string comparison.



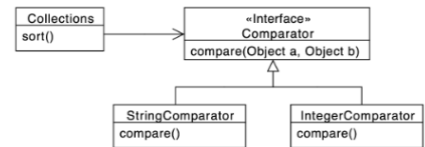
2.4.6 Strategy Pattern

Many classes or algorithm has a stable core and several behavioral variations, several different implementations are needed. Every variation of an algorithm can be embedded in an object and passed as a parameter to the algorithm.

Example: Comparator in Java.

Consequences:

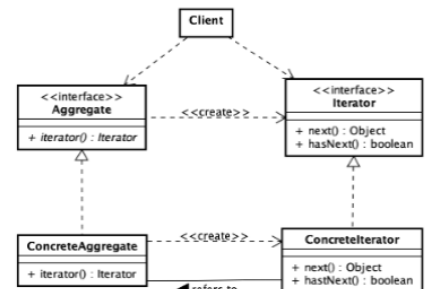
- + Avoid conditional statements
- + Algorithms may be organized in families
 - + Choice of implementations
 - + Run-time binding
- Clients must be aware of different strategies
 - Communication overhead
 - Increased number of objects



2.4.7 Iterator pattern

You have a collection of objects and you want to iterate on it allowing multiple concurrent iterators.

Example: In Java there is the Iterable interface and Iterator.



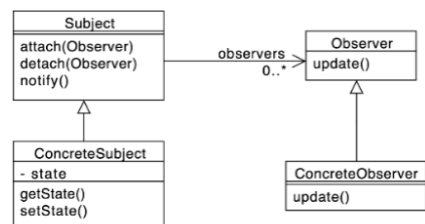
2.4.8 Observer pattern

The change in one object may influence one or more other objects. This can cause high coupling between objects and the number and type of objects to be notified may not be known in advance.

Example: In Java there is the Observable interface and Observer interface. This is very common in UI programming.

Consequences:

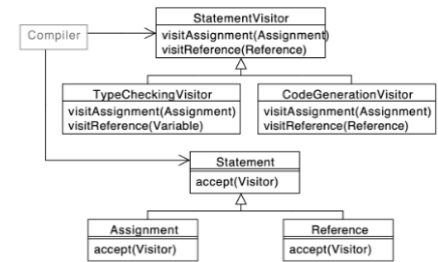
- + Abstract coupling between Subject and Observer
- + Support for broadcast communication
 - Unanticipated updates



2.4.9 Visitor

You have an object structure that contains many classes with different interfaces and you want to perform several different and unrelated operations on these objects.

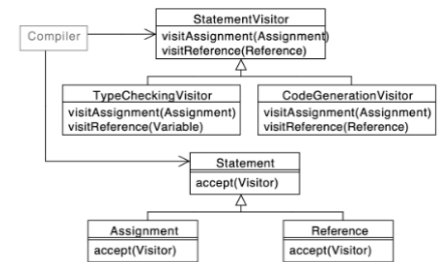
The operations depend on the concrete classes and you want to avoid putting many different methods in each of these classes.



2.5 Inheritance vs. composition

Reuse can be achieved via:

- **Inheritance:** a new class is created by inheriting from an existing class and adding new features or modifying existing ones. Clients can invoke directly inherited methods.
- **Composition:** The class has a link or instance to a class you want to reuse and just forwards the calls to it.



Note: The Observable pattern is deprecated. The PropertyChangeSupport is more flexible.

2.6 Abstract Syntax Tree

The AST is a tree that represent a syntax of formal language. We can use the composite pattern to representation the arithmetic expressions.

And using the visitor pattern or the interpreter pattern to evaluate the expression.

2.7 Model View Controller

Capitolo 3

What is software architecture ?

The architecture of a software system is the shape given to that system by those who build it. The form of that shape is in the division of that system into components, the arrangement of those components, and the ways in which those components communicate with each other. The purpose of that shape is to facilitate the development, deployment, operation, and maintenance of the software system contained within it.

3.1 Goals of an architecture

The software must change to adapt to software regulation, user feedback.

We have some pattern that make methods and classes easy to be changed, but now we want something behind the methods.

The goals of an architecture are:

- keep the system valuable for its users over time
- minimize the lifetime cost of the system
- maximize programmer productivity

This can be achieved by making the system:

- easy to understand
- easy to develop
- easy to maintain
- easy to deploy
- easy to change

In Java we have the package, that is a module, and is a logical grouping of classes.

Design Patterns: helps to build methods and classes inside the module in a way that they are easy to be changed.

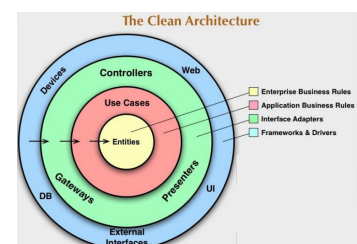
Design Principles: define the modules can collaborate together and how they are shaped.

3.2 The clear architectures

The **Outer circles:** mechanisms, they change often.

The **Inner circles:** policies, subject to less frequent changes.

Source code dependencies point only inward, toward higher-level policies (Dependency Rule)



3.2.1 Entities

An Entity is an object within a software system that contains a set of critical business rules operating on Critical Business Data.

The interface of the Entity consists of the functions that implement the Critical Business Rules to operate on the Critical Business Data. They contain the most general and highlevel rules and neither they are often changed nor they are usually affected by lower-level changes.

Example: The system is the university portal and an important entity is the student, that has some critical business rules like "the student can enroll to the master degree only if they have completed the required prerequisites and then assign new ID".

Is the most stable part of the system and we want to preserve as much as possible from changes occurring to the other elements.

Student
- ID - DateBirth - Address
+ assignGrade() + assignSubsidie() + computeTax()

3.3 Use cases

To apply the Business Roles to the entities we need to use the use cases.

A use case is a description of the software behavior and interactions (with people or other software) in a specific case.

It specifies:

- the input to be provided by the user (or other systems)
- the output to be returned
- The processing steps involved.

to determine the flow of data to and from the entities.

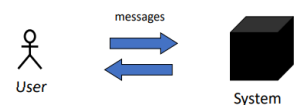
Software at this level change whenever a change to any element of a use case is required

Use case in pratice

It represents a **functional requirement**.

- It makes clear what you expect from a system ("what?")
- It hides the behaviour of the system ("how?")
- It is a sequence of actions (with variants) that produce a result observable by an actor

The system is seen as a black-box. Use cases express the interaction (exchange of messages) between the entities that interact with the system itself (called actors).



Example - Specify annual tax for a student Input: student ID

Output: annual tax amount

Main scenario:

1. Validate ID provided by user
2. Retrieve economic level
3. Compute due past taxes
4. Compute new taxes
5. Get average grades for the last academic year
6. If average grade > 26, apply 20% discount
7. Ask operator for confirmation: if yes change student else reset

3.4 Interface adapters

Their goal is to shape data in a way that it is convenient to higher levels or other systems or artifacts interacting with:

The same is done in the opposite direction. Examples of code in this layer: Presenters for GUI, Controller for DB

Code inward to this level know anything about details (e.g., SQL)

3.5 Frameworks and drivers

In order to achieve the separation we have seen so far, we need to separate low level details from the high-level details. Layers allow independent development, deployment, operation, and maintenance.

The details are the frameworks, the database, the drivers, the web server, etc., and usually we do not write much code here.

3.6 Separation of policy from details

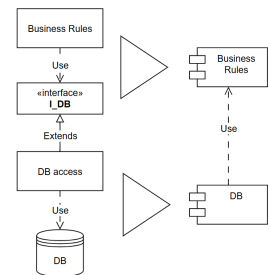
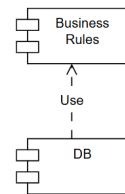
Decoupling of policy from the details and defer decisions on details for as long as possible (MongoDB or SQL, etc.)

DB component knows about use case rules component, but NOT viceversa.

Low level components know everything of the higher-level components, but NOT viceversa

The Database component contains the code that translates the calls made by the BusinessRules into the query language of the database. This translation code knows about the BusinessRules.

Business rules can be adapted to any kind of DB.



Capitolo 4

From design to architecture: Design Principles

Design principles means follow a clean architecture and a clean architecture means that the architecture can easily evolve, to be changed.

4.1 The SOLID design principles

The SOLID principles suggest software architects on how to arrange functions and data into groupings (e.g., classes), and how these groupings are arranged.

The goal of the SOLID principles is to create mid-level software structures (modules) that:

- Are easy to understand
- Allow changes in the most "comfortable" way
- Are the basis of reusable components

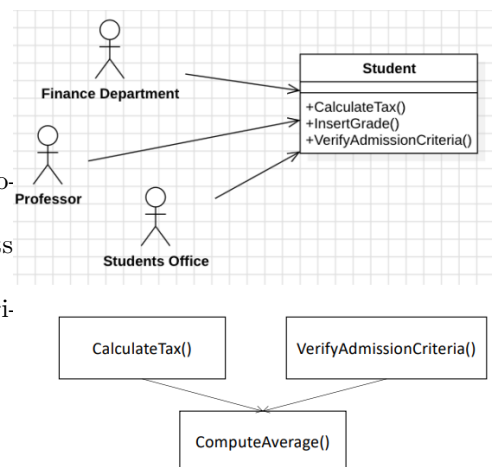
4.1.1 Single Responsibility Principle - SRP

A module should have one, and only one, reason to change. It means that a module should be responsible to **one**, and only one, **actor**.

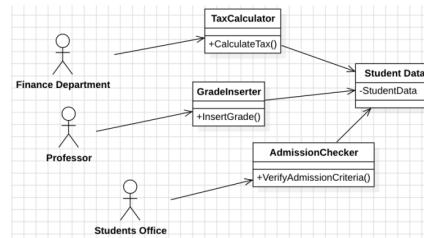
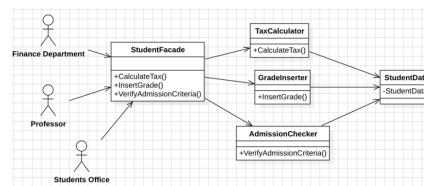
This is helpful to separate code used by different actors and for a better maintenance.

Example

CalculateTax() is specified by the Finance Department.
InsertGrade() is specified by Students Office and used by Professor
VerifyAdmissionCriteria() is specified and used by Students Office
CalculateTax() and *VerifyAdmissionCriteria()* share a common algorithm for calculating the average grade.



Changes in *ComputeAverage()* affect two actors: errors may arise and also changes to Class Student due to different needs of actors may lead to collisions and merges.

Solution 1: Separate data from functions**Solution 2: Use the Façade pattern**

You increase coupling and complexity, its a trade off. But the separation is splitted.

Note: You increase coupling and complexity but the separation is splitted.

Solution 2: variation**4.2 Open-Closed Principle - OCP**

A software artifact should be **open for extension but closed for modification**. Using design patterns to make new functionality in a way that is additive and not to modify the existing structure.

The behavior of a software artifact ought to be extendible, without having to modify that artifact.

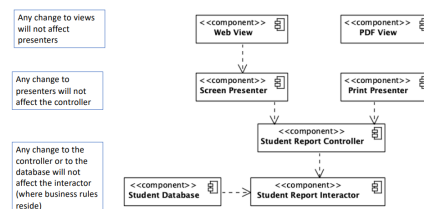
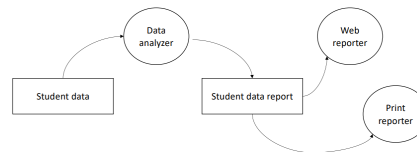
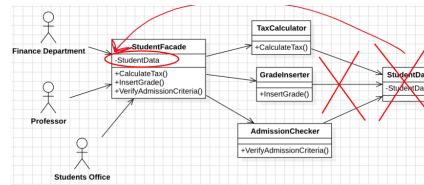
The aim of OCP is to make the system easy to extend without requiring too many changes to the system.

Using the open-closed principle, the software architect divides the system into components and arranges these components in a dependency hierarchy that protects higher-level components from changes in lower-level components.

Example A system displays summary information about a student's career at the university on a dashboard: Subjects (those passed with grades), Taxes paid and due, Current semester schedule, Next exams, Important notices, Stakeholders ask for a printed report with only information on subjects and taxes, and a different formatting.

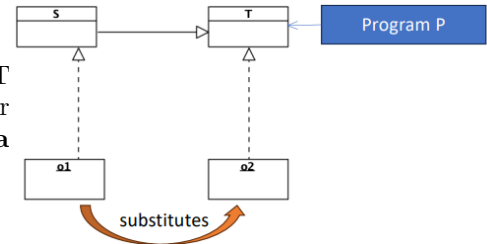
Generating the report involves two separate responsibilities:

- the calculation of the reported data
- the presentation of that data into a web- and printer-friendly form.



4.2.1 Liskov substitution principle - LSP

If for each object *o1* of type *S* there is an object *o2* of type *T* such that for all programs *P* defined in terms of *T*, the behavior of *P* is unchanged when *o1* is substituted for *o2* then ***S* is a subtype of *T***.

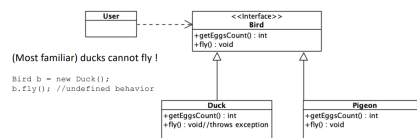


If we substitute *o2* with *o1* and the behavior of *p* does not change means *S* is derived class of *T*.

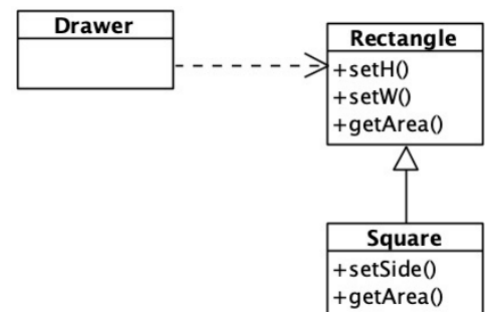
This principle is about subtyping, not only for inheritance or extension but also for the logic, the behavior of the class.

Derived type must fully support the substitution of their base types: functions that use pointers or references to base classes must be able to use objects of the derived without knowing it. This is related to the substitution property.

A subtype can replace its supertype only if it behaves the same way. This idea, called **behavioral subtyping**, means the subtype doesn't just have the same methods as the supertype — it also follows the same rules for how those methods should work. That way, anything the client expects from the supertype will also hold true for the subtype.



```
Rectangle r = ...;
r.setW(5); r.setH(2);
assert(r.area() == 10);
```



Depending on how *r* is defined, the assertion can fail because you not use the *setH()* and *setW()* methods of the rectangle but you use the *setSide()* method of the square.

How to refactor it? Using the Interface segregation principle - ISP

4.2.2 Interface segregation principle - ISP

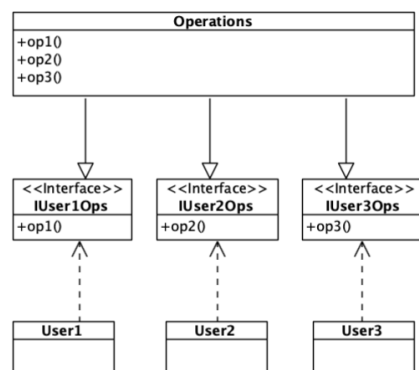
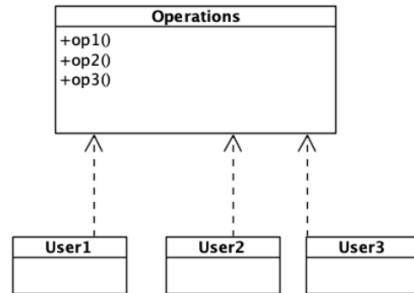
Clients should not be forced to depend on operations and/or interfaces they do not use. Better to make fine grained interfaces that are client specific.

Similar to SRP we have different actors, represented by different classes. They are only interested in a subset of the operations, so we can split the interface in *U1* use only operation1 and *U2* use only operation2 and *U3* use only operation3

In a statically typed language (e.g., Java), any change to *op1()* will force recompilation and redeployment of *User2* and *User3*.

Is different from the SRP because we need the classes but we need to modularize better.

Inserting interfaces the classes use the interfaces instead the class and the interface implement the class.

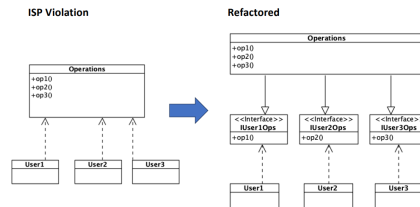


If we change operation1 we impact also user2 and user3, if we change the interface we impact only the corresponded user.

We also package the interface and the user together and this not impact the module.

Operations are segregated into interfaces.

In a statically typed language (e.g., Java), a change to *op1()* will NOT force recompilation and redeployment of User2 and User3.



Remove useless operations from modules, otherwise they will force useless redeployments

4.2.3 Dependency inversion principle - DIP

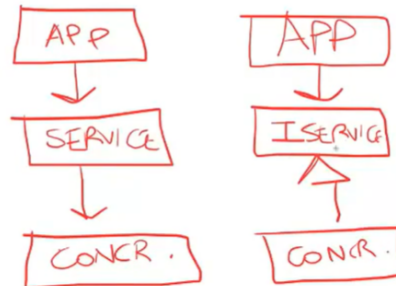
You should always have dependencies that lower modules depend on higher modules, not the opposite.

High-level modules should not depend on anything from low-level modules.

Stable abstractions: abstractions should not depend on details, so that changes to details will not affect them.

Idealistic tension, in practice each system has at least a concrete component: the main function

We can make an interface:



Also we can package the app and the interface together.

4.2.4 DIP: practical implications

Dependencies on concretions are tolerable when it comes to operating systems, platform facilities and other similar components whose changes are tightly controlled and not frequent.

Abstract Factories are helpful to adhere to the DIP by minimizing the number of dependencies on concrete implementations and keep them separate from the rest of the system

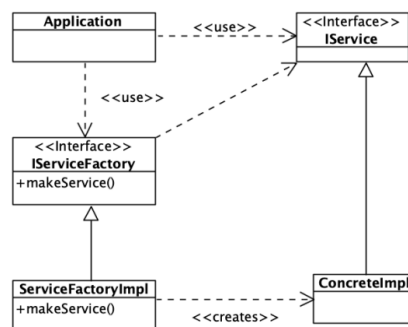
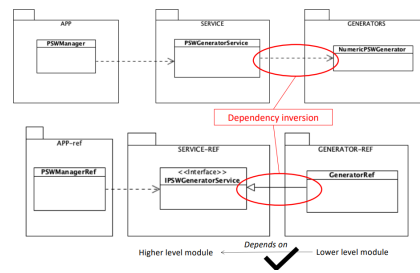
In general, volatility should be shifted towards implementation details without changing interfaces.

Using Factory Pattern

To create an instance of *ConcreteImpl*, the Application calls *makeService()* from *IServiceFactory* interface, which is implemented by *ServiceFactoryImpl* class.

ConcreteImpl is created and returned as a *IService*.

Separation abstract and concrete levels: All source code dependencies point toward the abstract level. The flow of control crosses the separation line in the opposite direction: this is the sense of "dependency inversion".



DIP violation DIP violations cannot be entirely removed. Some dependencies on concrete implementations are necessary. They must be minimized and kept separate